Ex. 5

Q2 Goal $P(HD = Yes \mid E = Yes, BP = High, CP = Yes)$

$= P(HD = Yes \mid BP = High, CP = Yes, E = Yes)$

$= \dfrac{P(BP = High, CP = Yes \mid HD = Yes, E = Yes) \times P(HD = Yes \mid E = Yes)}{P(BP = High, CP = Yes \mid E = Yes)}$

$= \dfrac{P(BP = High, CP = Yes \mid HD = Yes) \times P(HD = Yes \mid E = Yes)}{P(BP = High, CP = Yes \mid E = Yes)}$

$= \dfrac{P(BP = High \mid HD = Yes) \times P(CP = Yes \mid HD = Yes) \times P(HD = Yes \mid E = Yes)}{P(BP = High, CP = Yes \mid E = Yes)}$

$= \dfrac{0.85 \times 0.8 \times 0.25}{P(BP = High, CP = Yes \mid E = Yes)}$

$= 0.65$

$P(HD = No \mid E = Yes, BP = High, CP = Yes)$

$= 1 - 0.65 = 0.35$

Concl: It is likely that this patient/person will suffer from Heart Disease

$P(BP=High, CP=Yes \mid E=Yes)$

$= P(BP=High, CP=Yes \mid E=Yes)$

$= P(BP=High, CP=Yes \mid HD=Yes, E=Yes) \times P(HD=Yes \mid E=Yes)$

$+ P(BP=High, CP=Yes \mid HD=No, E=Yes) \times P(HD=No \mid E=Yes)$

$= P(BP=High, CP=Yes \mid HD=Yes) \times P(HD=Yes \mid E=Yes) +$

$P(BP=High, CP=Yes \mid HD=No) \times P(HD=No \mid E=Yes)$

$= P(BP=High \mid HD=Yes) \times P(CP=Yes \mid HD=Yes) \times P(HD=Yes \mid E=Yes)$

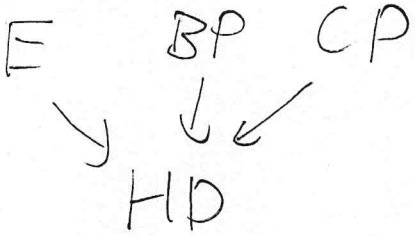
$+ P(BP=High \mid HD=No) \times P(CP=Yes \mid HD=No) \times P(HD=No \mid E=Yes)$

$= 0.85 \times 0.8 \times 0.25$

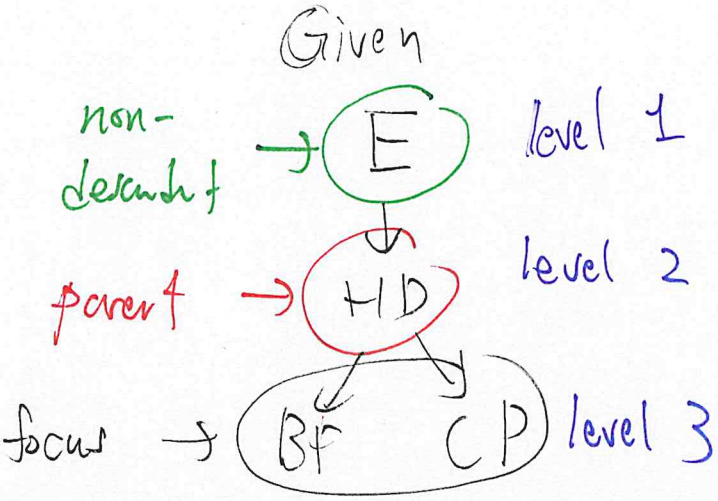
$+ 0.2 \times 0.6 \times (1-0.25)$

$= 11$

Goal


E
BP
CP
HD

Given


non-descendent
E
level 1
parent
HD
level 2
focus
BP
CP
level 3

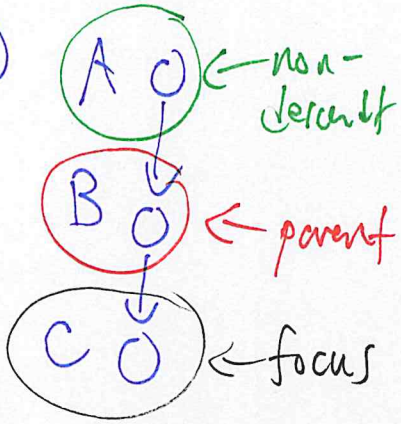
① $P(A|B,E,F) = \dfrac{P(B|A,E,F)\,P(A|E,F)}{P(B|E,F)}$

② $P(C,D|E,F)$

$= P(C,D|\text{Case 1},E,F) \times P(\text{Case 1}|E,F)$

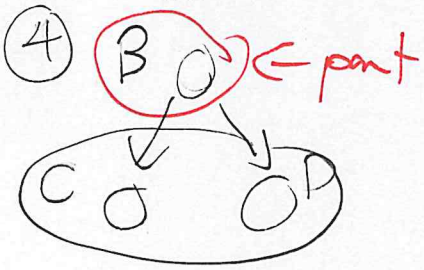
$+ P(C,D|\text{Case 2},E,F) \times P(\text{Case 2}|E,F)$

③



$P(C|A,B) = P(C|B)$

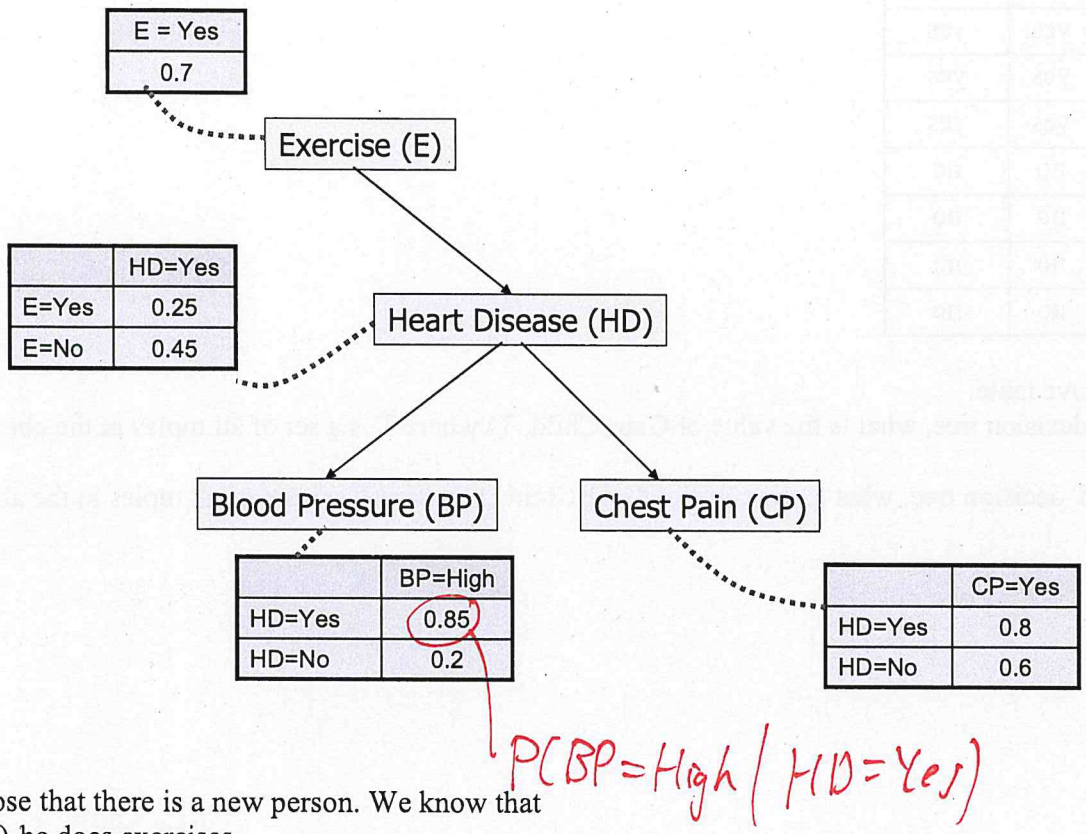
④



$P(C,D|B) = P(C|B) \times P(D|B)$

Q2 Bayes Classifier

We have the following Bayesian Belief Network.



Suppose that there is a new person. We know that

(1) he does exercises
(2) he has high blood pressure
(3) he has chest pain

We would like to know whether he is likely to have Heart Disease.

Exercise	Blood Pressure	Chest Pain	Heart Disease
Yes	High	Yes	?

Please use Bayesian classifier with the use of Bayesian Belief Network to predict whether he is likely to have Heart Disease.

$$3x_1 + 4x_2 - 12 = 0$$

LHS $= 3 \times 12 + 4 \times 13 - 12$

> 0

LHS $= 3 \times (+ 4 \times 2 - 12$

< 0

$$w_1 x_1 + w_2 x_2 + b - D = 0$$

$$\frac{w_1}{D} x_1 + \frac{w_2}{D} x_2 + \frac{b}{D} - 1 = 0$$

w_1' w_2' b'

$$w_1 x_1 + w_2 x_2 + b - 1 \geq 0$$

$$w_1 x_1 + w_2 x_2 + b \geq 1$$

$$(+1)(w_1 x_1 + w_2 x_2 + b) \geq 1 \cdot (+1)$$

$$y(w_1 x_1 + w_2 x_2 + b) \geq 1$$

$$w_1 x_1 + w_2 x_2 + b + 1 \leq 0$$

$$w_1 x_1 + w_2 x_2 + b \leq -1$$

$$(-1)(w_1 x_1 + w_2 x_2 + b) \geq (-1) \cdot (-1)$$

$$y(w_1 x_1 + w_2 x_2 + b) \geq 1$$

$$y(w_1 x_1 + w_2 x_2 + b) \geq 1$$

$$(+1)(w_1 \cdot 12 + w_2 \cdot 13 + b) \geq 1$$

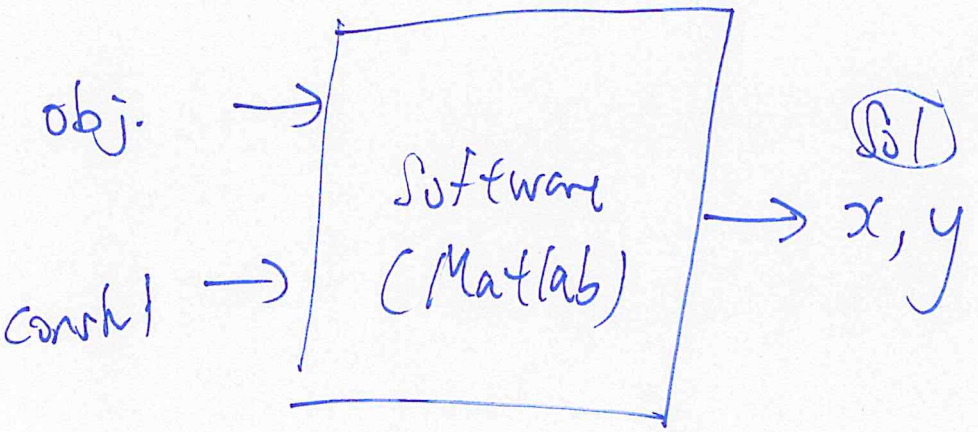
$$12 w_1 + 13 w_2 + b \geq 1 \quad \leftarrow \text{linear}$$

$$(-1)(w_1 \cdot 1 + w_2 \cdot 2 + b) \geq 1$$

$$-w_1 - 2 w_2 - b \geq 1 \quad \leftarrow \text{linear}$$

Constrained Programming

	Linear Programming	Quadratic Progrmg	Non-liner Progrmg
objective -functn	linear	quadratic	non-liner
constraints	liner	linear	non-line



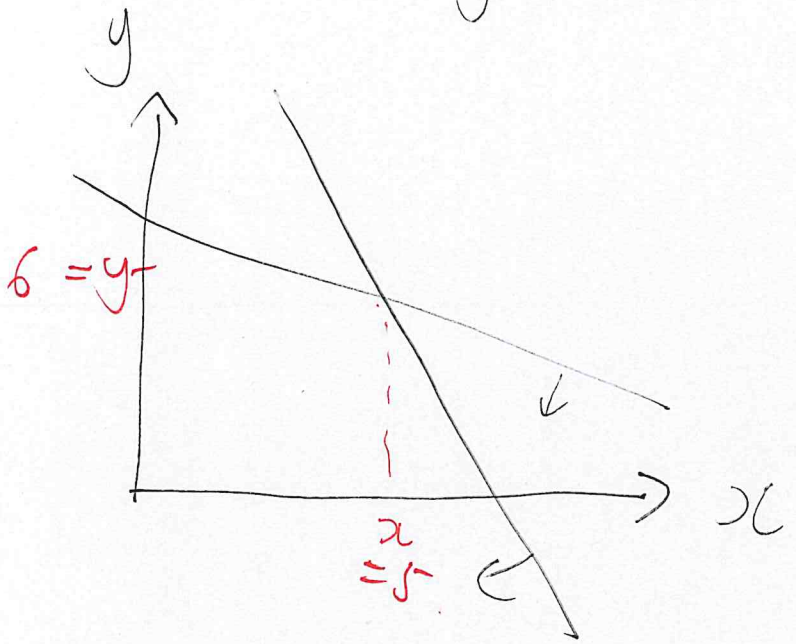
max. $x + 7y$ ← linear

s.t.

$2x + 5y \leq 10$ ← linear

$3x + 2y \leq 7$ ← linear

$x, y \in \mathbb{R}$



max $x^2 + \sqrt{\dfrac{xy + \sqrt{x}}{\cos x}}$